

YIYUAN, China

WMO: 548360

Lat: 36.183N

Lon: 118.150E

Elev: 302

StdP: 97.75

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.9	(c) -8.3	(d) -23.9	(e) 0.4	(f) -3.6	(g) -21.6	(h) 0.6	(i) -3.3	(j) 7.0	(k) -1.9	(l) 6.1	(m) -0.9	(n) 1.1	(o) 250	(p) 0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	33.4	23.1	32.2	22.7	31.0	22.4	26.7	30.3	26.0	29.5	25.2	28.5	3.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.8	21.8	28.8	25.0	20.8	28.2	24.2	19.9	27.6	86.2	30.4	82.7	29.5	79.4	28.5	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.6	(b) 5.6	(c) 5.0	(e) -12.8	(f) 35.5	(g) 2.2	(h) 1.5	(i) -14.4	(j) 36.6	(k) -15.7	(l) 37.5	(m) -17.0	(n) 38.4	(o) -18.6	(p) 39.5		
(a) 6.6	(b) 5.6	(c) 5.0	(e) -13.9	(f) 27.5	(g) 2.1	(h) 0.8	(i) -15.5	(j) 28.1	(k) -16.7	(l) 28.6	(m) -17.9	(n) 29.0	(o) -19.4	(p) 29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	-1.6	1.3	7.5	14.4	20.0	24.0	26.3	25.2	20.7	14.6	6.9	0.5	(6)
(7)		DBStd	10.28	3.27	3.77	4.38	4.14	3.30	2.58	2.32	2.48	2.85	3.37	4.00	3.36	(7)
(8)		HDD10.0	1121	359	243	104	9	0	0	0	0	0	5	107	293	(8)
(9)		HDD18.3	2608	618	477	337	131	20	1	0	0	11	122	342	552	(9)
(10)		CDD10.0	2355	0	0	26	139	310	419	506	470	321	148	16	0	(10)
(11)		CDD18.3	802	0	0	1	12	72	171	248	212	81	6	0	0	(11)
(12)		CDH23.3	7026	0	0	6	119	683	1538	2377	1762	499	42	0	0	(12)
(13)		CDH26.7	2366	0	0	1	24	200	579	899	554	106	3	0	0	(13)
(14)	Wind	WSAvg	2.0	1.8	2.1	2.5	2.7	2.5	2.1	2.0	1.9	1.7	1.7	1.8	1.8	(14)
(15)	Precipitation	PrecAvg	701	8	10	17	35	46	89	227	139	71	31	16	9	(15)
(16)		PrecMax	1617	40	38	48	115	144	258	480	460	332	86	75	57	(16)
(17)		PrecMin	408	0	0	1	0	3	4	30	32	10	0	0	0	(17)
(18)		PrecStd	238	10	11	12	27	35	60	110	88	62	25	16	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	14.8	23.7	29.6	33.2	34.7	35.1	33.9	31.2	26.5	20.7	12.1	(19)
(20)			MCWB	3.4	8.1	13.2	17.7	17.6	20.6	24.1	25.7	21.0	15.9	12.0	7.1	(20)
(21)		2%	DB	7.5	12.0	20.4	26.4	30.7	32.9	33.6	32.1	29.2	24.5	17.7	9.7	(21)
(22)			MCWB	2.2	5.6	10.6	15.5	17.5	19.9	24.6	25.0	20.2	15.2	10.2	5.5	(22)
(23)		5%	DB	5.7	10.1	17.7	24.3	28.8	31.5	32.4	30.8	27.7	22.6	15.5	7.7	(23)
(24)			MCWB	1.2	4.3	9.4	14.2	16.8	20.0	24.6	24.2	19.4	14.7	9.4	3.6	(24)
(25)		10%	DB	4.0	8.3	15.2	22.1	27.1	30.2	31.0	29.7	26.3	20.8	13.5	6.0	(25)
(26)			MCWB	0.1	3.0	7.9	13.0	16.2	19.7	24.3	23.6	19.0	14.1	8.7	2.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	9.6	14.5	19.6	21.6	25.0	27.6	27.7	24.5	19.3	14.2	9.1	(27)
(28)			MCDWB	7.9	12.7	21.1	26.1	26.0	29.9	31.6	31.3	27.9	21.4	16.9	10.8	(28)
(29)		2%	WB	3.2	6.8	12.1	17.2	20.0	23.8	26.8	26.7	22.6	17.6	12.3	6.0	(29)
(30)			MCDWB	6.2	11.2	18.5	23.9	26.1	28.3	30.6	30.2	25.8	21.0	15.2	8.2	(30)
(31)		5%	WB	1.6	5.0	10.2	15.7	19.0	22.7	26.2	25.9	21.6	16.4	10.7	4.3	(31)
(32)			MCDWB	5.1	8.8	16.5	22.2	25.6	27.3	30.0	29.1	25.0	20.5	14.3	7.0	(32)
(33)		10%	WB	0.3	3.4	8.4	14.2	17.9	21.9	25.5	25.2	20.6	15.3	9.1	2.8	(33)
(34)			MCDWB	3.5	7.6	14.4	20.3	24.2	26.7	29.1	28.1	24.1	19.4	12.9	5.5	(34)
(35)	Mean Daily Temperature Range	MDBR	8.8	9.1	10.2	10.6	10.8	9.7	7.5	7.3	9.0	10.1	9.4	8.5	(35)	
(36)		5% DB	MCDBR	11.5	11.9	13.6	13.3	13.7	12.4	9.5	8.7	11.1	12.7	12.7	11.2	(36)
(37)			MCWBR	7.3	7.1	6.8	5.7	4.7	3.8	2.8	2.9	3.7	5.0	6.4	7.0	(37)
(38)		5% WB	MCDWB	10.1	10.7	12.7	12.2	10.8	8.6	7.5	6.9	8.0	9.3	10.4	9.5	(38)
(39)			MCWBR	6.8	7.1	6.9	6.2	5.1	3.8	3.0	2.9	3.6	4.7	6.6	6.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.499	0.569	0.599	0.618	0.635	0.758	0.730	0.682	0.618	0.630	0.564	0.503	(40)	
(41)		taud	1.769	1.663	1.634	1.639	1.644	1.470	1.574	1.646	1.717	1.626	1.701	1.786	(41)	
(42)		Ebn at Noon	636	643	676	696	694	611	625	643	655	583	568	597	(42)	
(43)		Edh at Noon	171	214	239	250	253	300	269	246	218	219	182	159	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.37	2.96	4.28	5.07	5.62	5.21	4.61	4.43	4.00	3.28	2.49	2.18	(44)	
(45)		RadStd	0.27	0.40	0.34	0.26	0.40	0.44	0.42	0.48	0.38	0.33	0.35	0.23	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page